

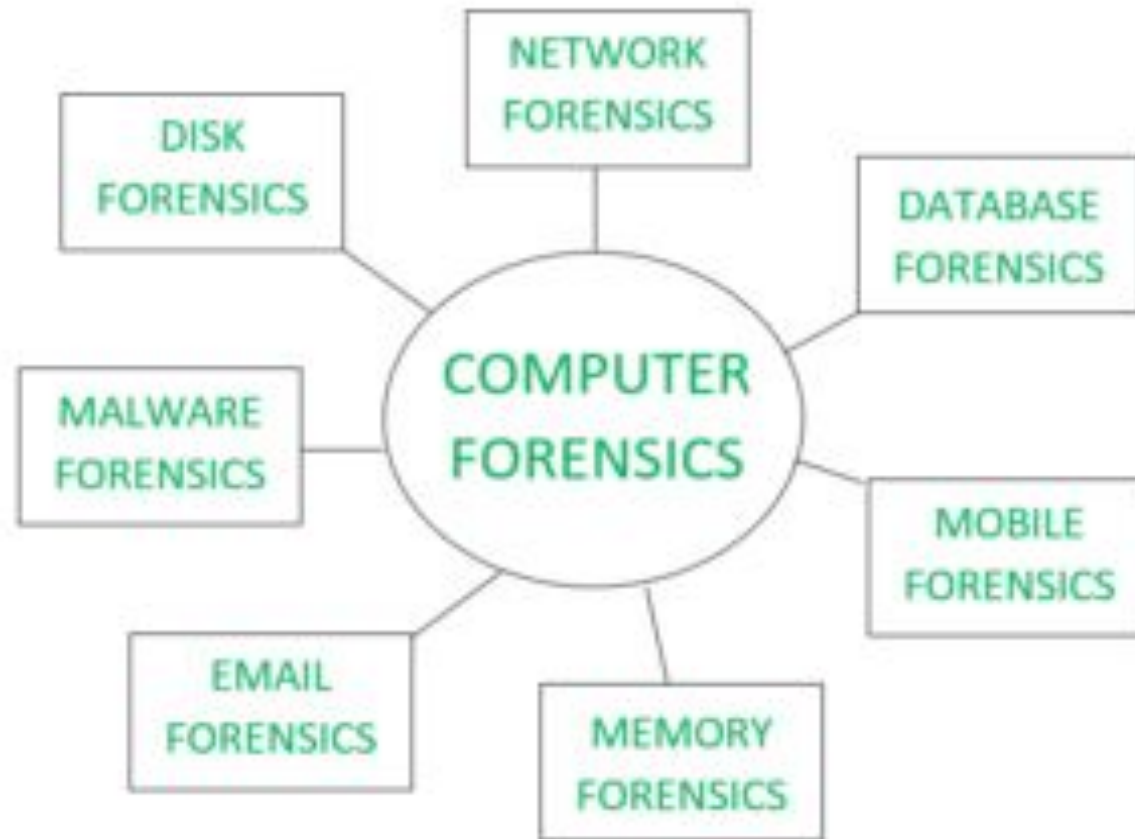
Computer Forensics

Module III

Introduction of Computer Forensics

- Investigative techniques to identify and store evidence from a computer device.
- Used to uncover evidence that could be used in a court of law.
- **Primarily used for two separate purposes:**
 - Investigation and
 - Data recovery.
- Computer forensics professional goes searching for data that is relevant to the investigation.
- When searching for information, they need to be careful to follow detailed procedures that allow their findings to be used as evidence.
- The information they uncover, whether it be documents, browsing information or even metadata, may then be used by prosecution attorneys to create a compelling case against the suspect.

Types of Computer Forensic



Disk Forensics

Analyzing storage devices such as hard drives, SSDs, USB drives, and memory cards to uncover digital evidence.

Whether you're investigating cybercrimes, recovering lost data, or ensuring compliance, mastering disk forensics is a vital skill for any forensic professional.

Disk Forensics Process

Identification : Identify relevant storage devices that may contain evidence.

Preservation :

Prevent data modification using **write blockers**.
Maintain **chain of custody**.

Acquisition (Imaging)

Create a **bit-by-bit forensic image** of the disk.
Hash values (MD5/SHA) are generated to verify integrity.

Analysis

Examine files, folders, file systems, slack space, and unallocated space.
Recover deleted files and analyze metadata.

Documentation & Reporting: Document findings clearly for court or organizational use.

Common Disk Forensic Techniques

File System Analysis – FAT, NTFS, EXT, APFS, etc.

Deleted File Recovery

Keyword Searching

Timeline Analysis

Hash Analysis

Data Carving (recovering files without file system info)

Tools Used in Disk Forensics

EnCase

FTK (Forensic Toolkit)

Autopsy / Sleuth Kit

X-Ways Forensics

Magnet AXIOM

dd / FTK Imager (for disk imaging)

Example

Case: An employee is suspected of stealing confidential company data.

Disk forensic investigation reveals:

- Deleted files recovered from unallocated space
- USB device usage logs
- Timestamps showing file copying before resignation

These findings become digital evidence in legal proceedings.

Memory Forensic

Deals with collecting data from system memory (system registers, cache, RAM) in raw form and then analyzing it for further investigation.

Focuses on identifying **live data** such as running processes, network connections, encryption keys, and malware that **do not exist on the hard disk**.

Objectives of Memory Forensic

Identify **running processes and services**

Detect **malware and rootkits**

Analyze **network connections**

Extract **encryption keys and passwords**

Reconstruct **system activity timeline**

Memory Forensics Process

Live Acquisition

Capture RAM while the system is running

Tools: DumpIt, FTK Imager (Live), Belkasoft RAM Capturer

Preservation

Save memory image securely

Generate hash values (MD5/SHA)

Analysis

Examine processes, DLLs, sockets, registry data

Identify suspicious or hidden activities

Documentation & Reporting: Record findings for legal use

Tools Used in Memory Forensics

Volatility Framework

Rekall

Magnet RAM Capture

Belkasoft RAM Capturer

FTK Imager (Live Capture)

Example Case

Scenario:

A company suspects a system is infected, but antivirus finds nothing on the hard disk.

Memory Forensic Investigation reveals:

A suspicious process running only in RAM

Malware injected into a legitimate process (e.g., `explorer.exe`)

Active communication with an external IP address

Encryption keys stored in memory

Result: **Fileless malware** confirmed through RAM analysis, which disk forensics alone could not detect.

Difference between Disk and Memory Forensic

Disk Forensics

Analyzes non-volatile data

System can be powered off

Files, logs, metadata

Evidence is permanent

Memory Forensics

Analyzes volatile data

System must be running

Processes, keys, connections

Evidence is temporary

Log forensics

Focuses on the **collection, examination, and analysis of log files** generated by operating systems, applications, servers, networks, and security devices.

Logs record **who did what, when, where, and how** on a system, making them crucial for reconstructing events during an investigation.

Objectives of Log Forensics

Track **user activities**

Reconstruct the **timeline of events**

Detect **security incidents and intrusions**

Identify **unauthorized access**

Provide **evidence** for legal and disciplinary actions

Types of Logs Analyzed

System logs (boot, shutdown, errors)

Security logs (login attempts, access control)

Network logs (firewalls, routers, IDS/IPS)

Web server logs (Apache, IIS)

Email logs

Application logs

Database logs

Log Forensics Process

Identification: Identify relevant log sources (OS, applications, network devices)

Collection:

- Securely collect log files without modification

- Use centralized log servers if available

Preservation:

- Maintain integrity using hash values

- Preserve chain of custody

Analysis:

- Correlate events across multiple logs

- Identify suspicious patterns or anomalies

Reporting: Document findings clearly and accurately

Registry Forensic

Suppose your computer lies in the hand of a malicious person without your consent.

Then how can you determine, what exactly he would have done to your computer.

You can track his activity through inspecting the registry.

Windows Registry is a hierarchical database that stores low-level settings for:

- The Windows operating system

- Hardware devices

- User profiles

- Applications

Evidence Can Be Found

User login details

Last login time

Recently opened files

USB devices connected

Installed/uninstalled software

Network connections

Malware persistence mechanisms

Windows Registry

HKEY_CURRENT_USER: loaded user profile for the currently logged-on-user.

HKEY_LOCAL_MACHINE: contains a vast configuration information for the system, including hardware settings and software settings.

HKEY_USERS: contains all the actively loaded user profile for that system

HKEY_CURRENT_CONFIG: contains the hardware profile the system uses at startup.

Windows Forensic Analysis

Windows Forensic Analysis focuses on 2 things:

1. In-depth analysis of Windows Operating System.
2. Analysis of Windows System Artifacts.

Windows artifacts are the objects which hold information about the activities that are performed by the Windows user.

The type of information and the location of the artifact varies from one operating system to another.

Windows artifacts contain sensitive information that is collected and analyzed at the time of forensic analysis.

Objectives of Windows Forensic

- Identify **what happened** on a Windows system
- Determine **who performed the action**
- Establish **when and how** the activity occurred
- Preserve evidence in a **legally admissible** manner

Types of Evidence in Windows Forensics

1. Volatile Data (Live Forensics):

Data that is lost when the system is powered off:

- Running processes

- Network connections

- Open files

- RAM contents

- Logged-in users

Example: Capturing RAM to find malware running only in memory

2. Non-Volatile Data (Dead Forensics):

Data stored on disk:

- Hard drives / SSDs
- System files
- Logs
- Registry data

Key Windows Forensic Artifacts

1. Windows Registry

The registry stores configuration and user activity data.

Important Registry Hives:

SAM – User account details

SYSTEM – System configuration

SOFTWARE – Installed programs

NTUSER.DAT – User-specific activities

Examples:

USB device history

Recently opened files

Installed applications

Continues

2. NTFS File System Artifacts:

Windows commonly uses **NTFS**.

Important NTFS Components:

- **MFT (Master File Table)** – Details of all files
- **\$LogFile** – File system transactions
- **\$UsnJrnl** – File creation, deletion, modification history

Example: Recovering deleted files and timestamps.

Continues

3. Windows Event Logs

Stored in `.evtx` files.

Common Logs:

- Security log – Logins, logouts, access attempts
- System log – System errors, shutdowns
- Application log – Application events

Example: Identifying unauthorized login attempts.

Continue

4. Prefetch Files

- Located in: `C:\Windows\Prefetch`
- Records information about executed programs

Example: Shows whether a program was executed and how many times.

Continue

5. Browser Artifacts

- History
- Cookies
- Cache
- Downloads

Example: Proving access to malicious or illegal websites.

Continue

6. Windows Recycle Bin

- Stores deleted files with original path and deletion time

Example: Recovering intentionally deleted evidence.

Continue

7. Memory (RAM) Forensics

- Detects rootkits, fileless malware, encryption keys
- Tools: Volatility, Rekall

Example: Extracting passwords or malware injected into memory.

Importance of Windows OS Forensics

- Cybercrime investigations
- Insider threat detection
- Malware analysis
- Legal and corporate compliance

Email forensics

Email forensics is dedicated to investigating, extracting, and analyzing emails to collect digital evidence as findings in order to crack crimes and certain incidents, in a forensically sound manner.

E-mail forensics refers to the study of email details including:

Email messages, Email addresses(sender and recipient), IP addresses, Date and time, User information, Attachments, Passwords logs (Cloud, server, and local computer)

Email Crimes

email crimes like

Homicide ,

Harassment,

cyber stalking,

cyber terrorism,

cyber threatening ,

cyber defamation,

fraud mails,

phishing,

email spamming ,

email bombing er

Email Forensics Works

Collection Phase: Investigators identify and preserve email evidence while adhering to strict chain-of-custody procedures.

includes capturing volatile memory during live incidents and securing stored email data across on-premises servers, cloud platforms, and mobile devices.

Examination Phase: Extract email artifacts using specialized tools and techniques.

Approach reveals the complete communication path and identifies potential spoofing or manipulation attempts.

Reporting Phase: Investigators analyze extracted data to identify patterns, authenticate communications, and establish timelines.

This involves correlating email metadata with other digital evidence, identifying suspicious communications patterns through behavioral analysis, and determining the scope of potential security incidents.

Reporting Phase: Teams document findings in formal reports that meet legal and compliance standards.

Reports include technical analysis results, investigative conclusions, and recommendations for remediation or further action, supporting both internal security improvements and external compliance requirements.

Introduction to Web Browser Forensics

Play vital part of any forensic investigation to determine what activity was carried out online, such as websites visited, searches performed and files downloaded.

Most web browsers provide the ability to create multiple browser profiles, with each profile storing history separately.

Therefore, browser history investigations may involve history from multiple profiles and multiple browsers.

We'll review some of the main desktop browser artifacts below:

We'll review some of the main desktop browser artifacts below:

1. Website Visits
2. Cache
3. Searches
4. Downloads
5. Timestamps
6. Chromium Apps
7. File Location and Formats
8. Browser Forensic Tools

Website Visits: By default web browsers log every website a user visits to a database on the user's computer.

How long this data is retained for depends on the browser and its settings

All browser's record the following information for each web page visited:

- Date/Time of the visit
- Title of the web page
- URL of the web page

It should be noted that if Private Browsing/Incognito mode is used then no website visits will be recorded by the browser.

It's also possible for the user to manually delete browsing history at any point.

Visit Type:

Chrome, Edge and Firefox record a "Visit Type", which describes how the user visited a web page.

The main visit types are:

- Link – the user clicked a link to the web page
- Typed – the user typed the URL into the address bar
- Bookmark – the user clicked a bookmark to the web page
- Other – the web page was visited by another means, such as a redirect

This can be useful for determining whether a user intentionally visited a particular web page or not.

Visit Source:

Another useful data point stored by Chrome and Edge is the "Visit Source" which describes if the website visit record was generated from somewhere other than the user navigating in the browser.

Visit sources can be:

- Synced – synchronised from another device
- Extension – added by an extension
- Imported – imported from another browser

Identifying if a website visit was synchronised from another device can be extremely important in providing context to the browser history.

Local Files

On a Windows computer, files that are opened via Windows Explorer get logged in the Internet Explorer history database (WebCacheV01.dat).

These records can be easily identified as the URL will start with file:///. Being able to review what files were opened by a user can be a useful addition to any investigation.

Cache

The cache is where web browsers temporarily store files from visited web pages, in order to load the web page faster on subsequent visits.

Cached Images: Images that are displayed on a web page get stored by the browser. Alongside the image file the browser will store metadata such as the URL of the image.

Cached HTML: Depending on the HTTP headers returned by a web page, the actual HTML of the web page may get stored by the browser

Searches: The phrases a user has been searching for online can be extremely useful in understanding the intent of the user's web browsing. For example, it's harder for a user to claim they accidentally clicked a link to a website, if they were actively searching for the particular topic of the website beforehand.

Download: All browser's record the following information for each file downloaded:

- URL the file was downloaded from
- Local path the file was saved to
- Date/Time of the download
- Number of bytes download or if the download was completed

Timestamps: Most browser history artifacts will have one or more timestamps associated with them.

These timestamps are usually stored as the units of time that have passed since a particular date (known as the epoch).

For example, a common format used is the number of seconds that have passed since 01/01/1970 00:00:00 (known as Unix time).

An overview of where the main artifacts are stored and in which format can be found for each browser below:

- [Chrome Forensics](#)
- [Edge Forensics](#)
- [Firefox Forensics](#)
- [Internet Explorer Forensics](#)
- [Safari Forensics](#)

Malware Forensics

way of finding, analyzing & investigating various properties of malware to seek out the culprits and reason for the attack.

The method also includes tasks like checking out the malicious code, determining its entry, method of propagation, impact on the system, ports it tries to use etc. investigators conduct forensic investigation using different techniques and tools.

Types of Malware

The category of malware is predicated upon different parameters like how it affects the system, functionality or the intent of the program, spreading mechanism, and whether the program asks for user's permission or consent before performing certain operations. a number of the commonly encountered malwares are:

- Backdoor
- Botnet
- Downloader
- Launcher
- Rootkit
- HackTool
- Rogue application
- Scareware
- Worm or Virus
- Credential-stealing program, etc.

4 Stages Of Malware Analysis

4 Stages of Malware Analysis:



Static Properties Analysis



Interactive Behavior Analysis



Fully Automated Analysis



Manual Code Reversing

Static properties analysis: Properties refer to strings of code embedded inside the malware file, hashes, header details, and metadata. Static properties analysis provides a quick and easy way to gather helpful information about malware because the malware does not have to be executed for you to study it.

Interactive behavior analysis: Analysis involves a security analyst interacting with malware running in a lab, making observations regarding its behavior. In this way, you can better understand how malware uses different elements of a computer system, such as its memory.

Fully automated analysis: Fully automated analysis scans suspected malware files using automated tools, focusing on what the malware can do once inside your system.

Manual code reversing: Reversing breaks down the code used to build the malware to learn how it works and what it is capable of doing.

Challenges in Computer Forensics

1. Rapidly Changing Technology: New operating systems, devices, apps, and storage methods appear frequently.

Investigators must constantly update their tools and skills.

2. Huge Volume of Data: Modern systems store terabytes of data

Finding relevant evidence is time-consuming and complex.

3. Encryption and Password Protection: Data is often protected using strong encryption.

Breaking encryption legally and technically is very difficult.

4. Anti-Forensic Techniques: Criminals use tools to hide, delete, or alter data (e.g., wiping software, steganography).

This makes evidence recovery harder.

5. Cloud Computing: Data may be stored across multiple cloud servers in different countries.

Jurisdiction and data access become complicated.

6. Legal and Jurisdiction Issues: Different countries have different cyber laws and privacy regulations.

Evidence collection must strictly follow legal procedures.

7. Maintaining Evidence Integrity: Evidence must not be altered during collection or analysis.

Even a small mistake can make evidence invalid in court.

8. Lack of Skilled Professionals: Computer forensics requires technical, legal, and analytical expertise.

There is a shortage of trained forensic experts.

9. Time Constraints: Investigations often need to be completed quickly.

Detailed analysis under time pressure increases difficulty.

10. Admissibility in Court: Evidence must be collected, analyzed, and documented properly.

Poor documentation can lead to rejection in court.

Tools used in Computer Forensics

1. EnCase

Widely used professional forensic tool

Helps in disk imaging, data recovery, and analysis

Accepted in courts as digital evidence

2. FTK (Forensic Toolkit)

Developed by AccessData

Used for disk analysis, email analysis, and password recovery

Fast indexing and searching of data

3. Autopsy

Open-source digital forensics tool

Used for analyzing hard drives, smartphones, and memory cards

Easy to use and suitable for beginners

4. The Sleuth Kit

Command-line forensic tool

Used to analyze file systems and recover deleted files

Often used with Autopsy

5. Wireshark

Network forensic tool

Captures and analyzes network traffic

Useful for detecting cyber attacks and data leaks

6. Cellebrite UFED

Used for mobile forensics

Extracts data from smartphones (calls, messages, apps)

Commonly used by law enforcement agencies

7. Volatility

Memory (RAM) forensic tool

Used to analyze running processes, malware, and system activity

Useful in live forensics

8. X-Ways Forensics

Lightweight and powerful forensic software

Used for disk imaging and file system analysis

9. Helix

Linux-based forensic live CD

Used for incident response and data acquisition

10. Magnet AXIOM

Used for computer, mobile, and cloud forensics

Recovers deleted files, chat messages, and browser data

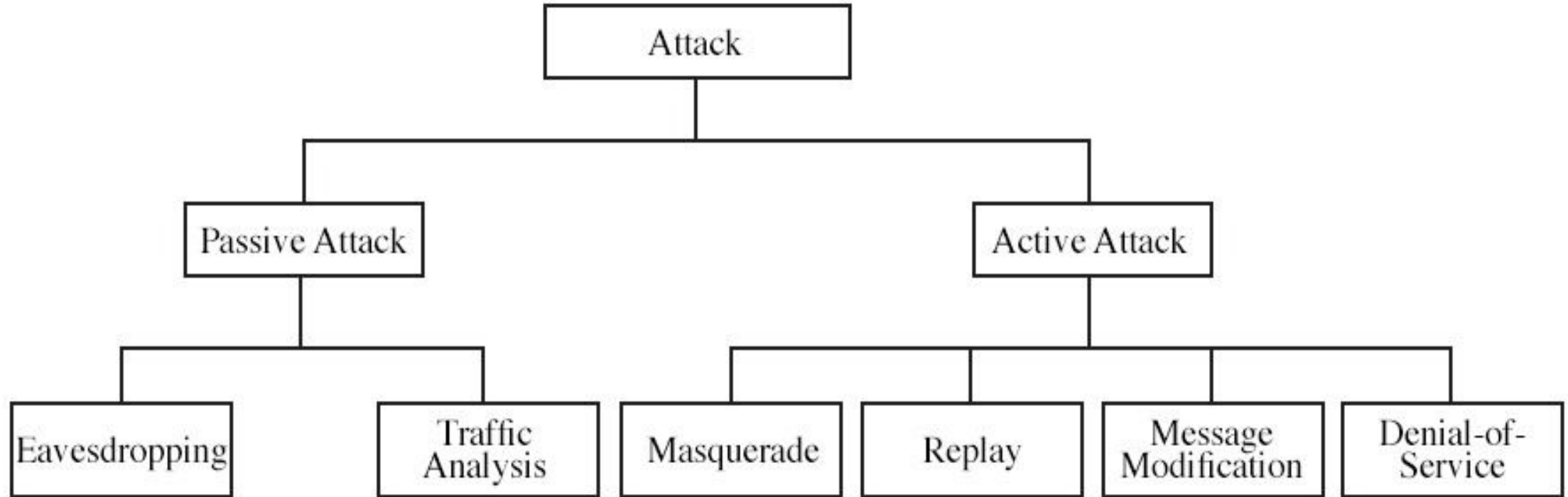
Security Attack

Illegal or unauthorized steps that put the security of a system at risk, it is known as a cyber security attack.

These kinds of attacks aim to corrupt, delete, or steal private data or information.

Mainly Cyber security attacks are of two types, i.e., active attack and passive attack.

Types of attacks

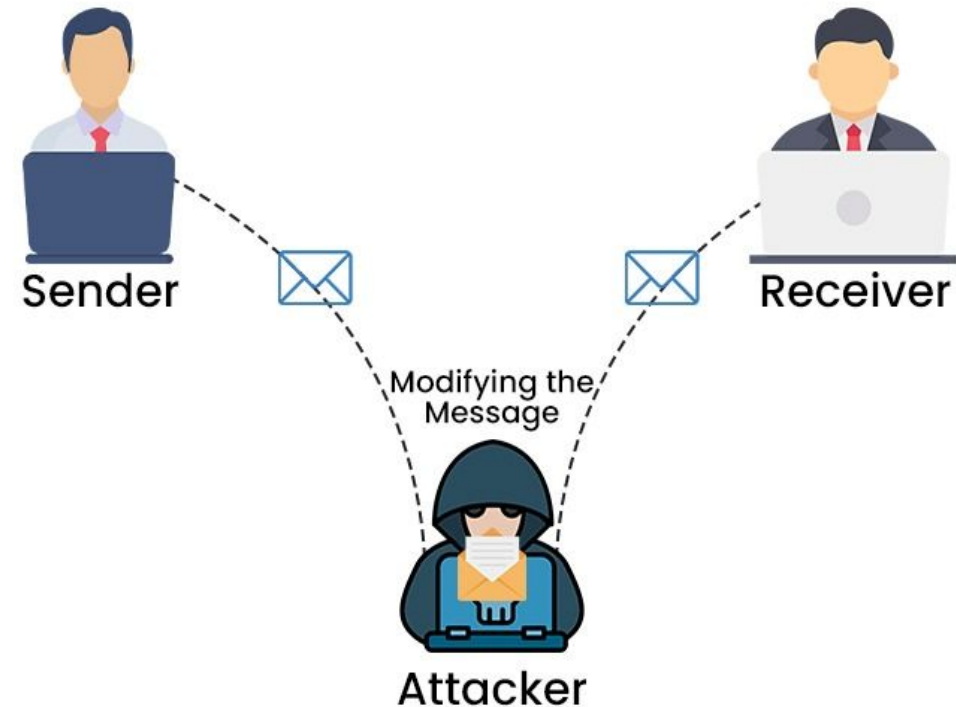


Active Attacks

Hacker or Attacker deliberately attempts to alter the original message or data.

Cause harm to systems and may even modify data.

Victim is alerted when an active attack is being carried out.



Types of Active Attacks

- **DOS attack:** It stands for Denial-of-Service attacks. Attacker floods the system with fake requests or data in order to make a service, website, or server unavailable. Types: DOS and DDOS
- **Message Modification:** Attacker modifies or delays the original information or data for its own benefit. For example, if someone sends the message “This message is for Rahul” and is confidential, the attacker might update the message to “This message is for Kevin.” By this, the attacker will get access to confidential information, which is only for Rahul.

- **Replay:** A replay attack refers to the act of capturing and transmitting a message with the intention of generating an authorized result.
With this attack, the attacker replicates the entirety of the data present within a network in order to utilize it for their own objectives in the future.
- **Masquerade:** In this type of attack, the attacker disguises himself and pretends to be a legitimate user.
With this, the attacker can easily steal sensitive information, which can cause many security issues for the user.

Defending against Active attacks

Here are a few measures to defend against active attacks:

- Install a firewall and intrusion detection/prevention systems.
- Use secure protocols like SSL/TLS and HTTPS.
- Install real-time monitoring tools to detect any threats to the system.

The following are a few preventive measures to take when facing such attacks:

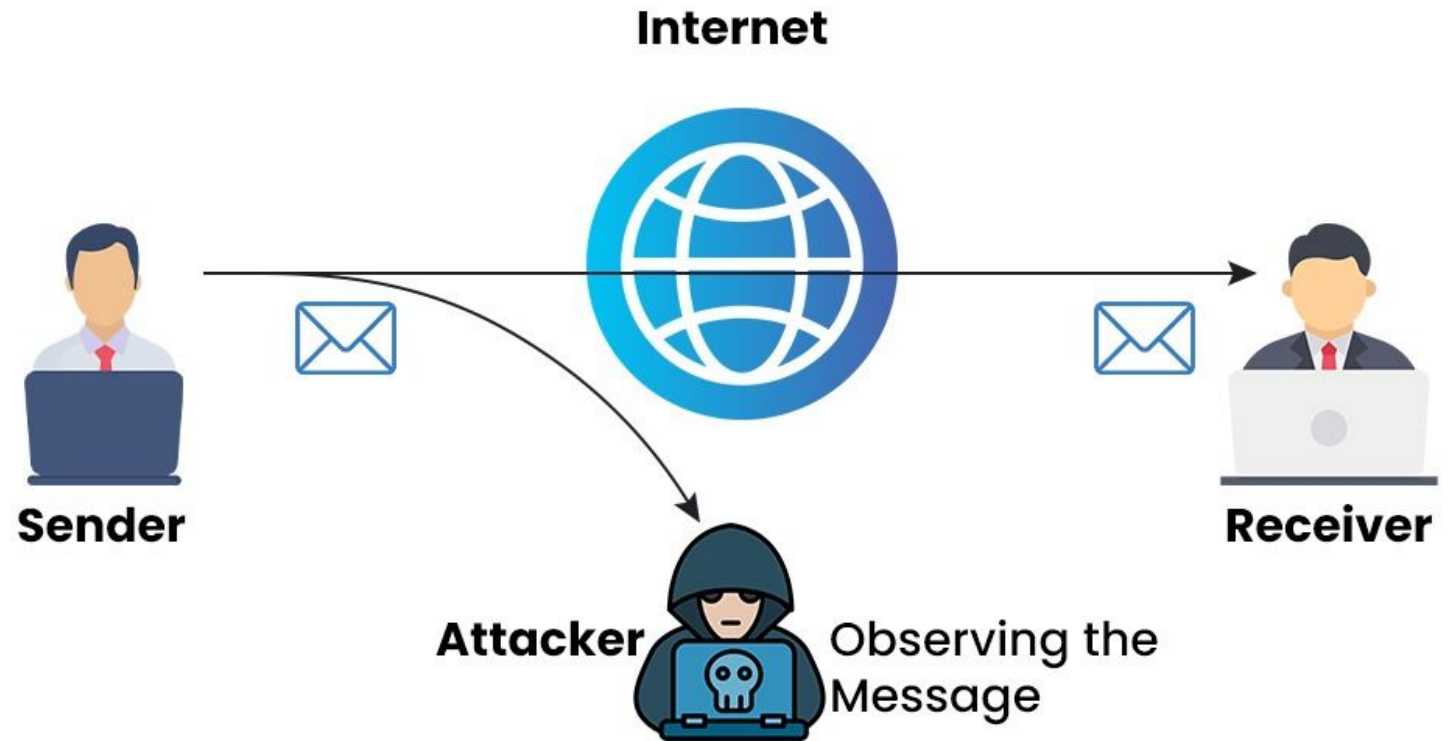
- Using one-time passwords is a helpful step in ensuring the authenticity between individuals.
- Another effective measure is to generate a random session key, ensuring that it is unique and exclusive to a single transaction. This will prevent any attacker from sending the altered message once the session is over.

Passive Attacks

attacker just monitors in on messages and steals any information they can get their hands on.

Monitoring every message and gathering information.

The attacker does not attempt to change the content of the information he steals



Passive Attack

Types of Passive Attacks

- **Eavesdropping:** In this type of attack, the attacker secretly listens to a conversation or communication that is not intended for him.
It can be done for various purposes, such as gathering information, spying, or blackmailing.
- **Traffic analysis:** Focuses on observing the exchange of information between two entities to extract valuable information.

Even the encrypted data can be used by attackers to discover the geographical location and the identities of the communicating entities, along with details regarding the frequency and duration of the exchanged messages.

This information is valuable in determining the nature of the ongoing communication.

Defending against Passive attacks

Here are some measures to be taken against passive attacks:

- You should encrypt all sensitive data with the latest encryption algorithms.
- Make sure your wireless systems are secure with strong passwords and WPA3.
- You should conduct regular security audits of your network to notice any passive attacks.

Below we have shown some of the preventive measures that one can take to prevent such attacks:

- If you don't want attackers to get access to your network, don't publish important information on social media.
- One can encrypt data in order to secure their personal information from attackers.